Readings

Sundaram: Chapter 5.1-5.5

Optimization with Equality Constraints

- The objective function is $f : \mathcal{R}^L \rightarrow \mathcal{R}$
- K $(< L)$ constraint functions,
  $g^1 : \mathcal{R}^L \rightarrow \mathcal{R}$, $g^2 : \mathcal{R}^L \rightarrow \mathcal{R}, \ldots,$ $g^K : \mathcal{R}^L \rightarrow \mathcal{R}$
  Note: $K < L$ makes the domain non trivial.

$$
\begin{array}{l}
\textit{maximize } f(\boldsymbol{x}) \\
\text{such that} \\
\boldsymbol{x} \in \mathcal{R}^L \\
g^1(\boldsymbol{x}) = 0, g^2(\boldsymbol{x}) = 0, \ldots, g^K(\boldsymbol{x}) = 0
\end{array}
$$

Local Maximum

Once again, we start from **local maximum** $x^\star$ that satisfy K constraints.

(i) $g^1(\boldsymbol{x}^\star) = 0, g^2(\boldsymbol{x}^\star) = 0, \ldots, g^K(\boldsymbol{x}^\star) = 0$

(ii) There exists a neighbourhood $B_\epsilon(\boldsymbol{x}^*)$ such that $f(\boldsymbol{x}^\star) \geq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in B_\epsilon(\boldsymbol{x}^*)$ and $g^1(\boldsymbol{x}) = 0, g^2(\boldsymbol{x}) = 0, \ldots, g^K(\boldsymbol{x}) = 0$.

Challenge: x^* is not an interior point.

First Order Condition :Theorem of Lagrange

Suppose that $f, g^1, \ldots, g^K$ are $\mathcal{C}^1$ functions.

$x^\star$ is local maximum (or minimum) of $f \Rightarrow$ There exists $\lambda_1^\star, \lambda_2^\star, \ldots, \lambda_K^\star \in \mathcal{R}^K$ such that

$$\nabla f(\boldsymbol{x}^\star) + \sum_{j=1}^{K} \lambda_j^\star \nabla g^j(\boldsymbol{x}^\star) = \mathbf{0}$$

$\lambda_1^\star, \lambda_2^\star, \ldots, \lambda_K^\star$ can be either positive or negative or zero.

First order conditions have L equations and $g^1(\boldsymbol{x}^\star) = 0, g^2(\boldsymbol{x}^\star) = 0, \ldots, g^K(\boldsymbol{x}^\star) = 0$ have K equations. These can be used to solve for $\boldsymbol{x}^*$ and $\lambda_1^\star, \lambda_2^\star, \ldots, \lambda_K^\star$.

First Order Condition :Theorem of Lagrange

The first order condition hold only when $rank(Dg(\boldsymbol{x}^\star)) = K$.

This condition is called *constraint qualification*.

$$Dg = \begin{bmatrix} \frac{\partial g^1}{\partial x_1} & \frac{\partial g^1}{\partial x_2} & \cdots & \frac{\partial g^1}{\partial x_L} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial g^K}{\partial x_1} & \frac{\partial g^K}{\partial x_2} & \cdots & \frac{\partial g^K}{\partial x_L} \end{bmatrix}$$

Classifying Criitical Points Using Second Order Condition

- If the following condition (involving the Hessian matrix) is satisfied at a critical point $\boldsymbol{x}^*$ then the critical point is interior local maximum (resp. interior local minimum)

$$\boldsymbol{z}^T \left[D^2 f(\boldsymbol{x}^\star) + \sum_{j=1}^{K} \lambda_j^\star D^2 g^j(\boldsymbol{x}^\star) \right] \boldsymbol{z} < 0 \text{ (resp. } > 0)$$

$$\text{for all } \boldsymbol{z} \in \mathcal{R}^L, \text{ such that } \boldsymbol{z} \neq \boldsymbol{0} \text{ and } Dg(\boldsymbol{x}^\star)\boldsymbol{z} = 0$$

- If the above condition holds with weak inequality then the critical point may or may not be interior local maximum (resp. interior local minimum). We need to find our own way to check it.

- If the above condition does not hold with either strong or weak inequality at a critical point then the critical point is neither interior local maximum nor interior local minimum.

Identifying Global Maximumor Minimum

- Step 1: Use Weierstrass' Theorem or some other argument to prove that global maximum (resp. minimum) exists.
- Step 2: Find all critical points using Therorem of Lagrange
- Step 3: Classify critical points using second order condition.
- Step 4: Compare $f(\boldsymbol{x})$ at all local maxima (resp. local minima).

Note: Usually constraint qualification hold at all points in the domain. If it does not there is no way to check whether CQ holds.

Caution

Caution 1: First order condition is a necessary condition. A critical point may not be a maximum/minimum.

$f(x_1, x_2) = x_1^3 + x_2^3$ and $g^1(x_1, x_2) = x_1 - x_2$. Note that CQ is met at all points.
First order condition:

$$\begin{aligned} 3x_1^2 + \lambda_1 &= 0 \\ 3x_2^2 - \lambda_1 &= 0 \\ x_1 - x_2 &= 0 \end{aligned}$$

Only solution is $x_1 = x_2 = \lambda = 0$. However $(0, 0)$ is neither local maximum nor local minimum. For instance, take
$f(\epsilon, \epsilon) = 2\epsilon^3 > 0 = f(0, 0)$, when $\epsilon > 0$.
This suggests that there is no global maximum and global minimum, which is indeed true. $f(\epsilon, \epsilon) = 2\epsilon^3$ goes to ∞ as $\epsilon \to \infty$

Caution

Caution 2: Lagrange thorem may fail because CQ is not satisfied.

Suppose $f(x_1, x_2) = -x_2$ and $g^1(x_1, x_2) = x_2^3 - x_1^2$. This has unique global maximum at $(0, 0)$. Note that CQ fails at $(0, 0)$. First order condition does not have any solution.

$$\begin{aligned} -2\lambda_1 x_1 &= 0 \\ -1 + 3\lambda_1 x_2^2 &= 0 \\ x_1^2 - x_2^3 &= 0 \end{aligned}$$